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ORIGINAL ARTICLE

# Body position influences arterial occlusion pressure: implications for the standardization of pressure during blood flow restricted exercise

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## Abstract

**Purpose** Arterial occlusion pressure (AOP) measured in a supine position is often used to set cuff pressures for blood flow restricted exercise (BFRE). However, supine AOP may not reflect seated or standing AOP, thus potentially influencing the degree of occlusion. The primary aim of the study was to investigate the effect of body position on AOP. A secondary aim was to investigate predictors of AOP using wide and narrow cuffs.

**Methods** Twenty-four subjects underwent measurements of thigh circumference, skinfold and blood pressure, followed by assessments of thigh AOP in supine and seated positions with a wide and a narrow cuff, respectively, using Doppler ultrasound.

**Results** In the supine position, AOP was 148 ± 19 and 348 ± 94 mmHg with the wide and narrow cuff, respectively. This increased to 177 ± 20 and 409 ± 101 mmHg in the seated position, with correlations between supine and seated AOP of $R^2 = 0.81$ and $R^2 = 0.50$ for the wide and narrow cuff, respectively. For both cuff widths, thigh circumference constituted the strongest predictor of AOP, with diastolic blood pressure explaining additional variance with the wide cuff. The predictive strength of these variables did not differ between body positions.

**Conclusion** Our results indicate that body position strongly influences lower limb AOP, especially with narrow cuffs, yielding very high AOP (≥ 500–600 mmHg) in some subjects. This should be taken into account in the standardization of cuff pressures used during BFRE to better control the physiological effects of BFRE.

**Keywords** Blood flow occlusion · Occlusion training · Pulse elimination pressure · Kaatsu · Arterial occlusion pressure

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### Abbreviations

| | |
|---|---|
| AOP | Arterial occlusion pressure |
| BFRE | Blood flow restricted exercise |
| BMI | Body mass index |
| DBP | Diastolic blood pressure |
| SBP | Systolic blood pressure |

## Introduction

Low-load blood flow restricted exercise (BFRE) has been shown to effectively improve muscle size and function, thus potentially providing a feasible resistance training alternative for individuals who may not tolerate the high mechanical load inherent of high-intensity resistance training (e.g., patients with musculoskeletal limitations) (Mattar et al. 2014; Tennent et al. 2016).

BFRE is typically performed with a pneumatic cuff placed proximally around the exercising limb and with

the cuff most often inflated to an absolute pressure aimed to partially reduce blood flow (Takarada et al. 2000). To accomplish this, several studies have employed empirically based absolute cuff pressures (Sumide et al. 2009; Nielsen et al. 2012) or relative cuff pressures based solely on brachial systolic blood pressure (Cook et al. 2007, 2017; Clark et al. 2011). However, standardization of a given degree of partial blood flow requires knowledge about the absolute cuff pressure needed to completely occlude arterial blood flow [i.e., arterial occlusion pressure (AOP)]. The challenge is that AOP is influenced by several factors such as cuff width, limb circumference and blood pressure (Crenshaw et al. 1988; Graham et al. 1993). In accordance, AOP has been reported to rise with increases in limb circumference and blood pressure (Graham et al. 1993; Loenneke et al. 2015), with relatively wide cuffs demonstrated to achieve AOP at lower absolute cuff pressures and to exhibit less variation than relatively narrow cuffs (Crenshaw et al. 1988; Loenneke et al. 2012). Standardization procedures using either fixed absolute pressures or pressures based solely on blood pressure do not take all of these factors into account. Consequently, such approaches will likely not succeed in sufficient equalization of the relative degree of arterial blood flow restriction (%AOP) between individuals. Therefore, the imposed BFRE stimulus will likely also differ, potentially confounding attempts to evaluate the adaptations with prolonged BFRE training. In recognition of this, previous studies have attempted to develop prediction equations for AOP of influencing factors (Graham et al. 1993; Noordin et al. 2009) and it has recently been proposed that such equations allow for improved individualized standardization of %AOP for BFRE (Loenneke et al. 2015).

An alternative approach to standardize the relative degree of blood flow restriction is to individually determine AOP by Doppler ultrasound, and then use a set percentage of AOP during BFRE (Laurentino et al. 2008, 2012). However, in previous studies using this approach, AOP has typically been determined with the subjects in a supine body position, while exercise of the lower extremities has often been performed in a seated body position (Laurentino et al. 2008; Fahs et al. 2015; Lixandrão et al. 2015). This approach does not consider the potential effect of postural changes on hydrostatic pressure (Folkow et al. 1971; Eiken 1988). Consequently, setting the cuff pressure based on AOP determined in a supine position before exercising in a different body position may result in overestimation of the %AOP applied during exercise. Furthermore, if AOP does not change homogeneously between subjects, lack of consideration of body position may introduce individual variation in the training stimulus. This especially applies if the BFRE regime is performed by a fixed repetition schemes (e.g., 30–15–15–15) often used in BFRE. In accordance, lack of control of individual variation in AOP may result in different levels of fatigue. This may prove important as factors inherent of fatigue development has been suggested to constitute important drivers of muscle accretion (Farup et al. 2015).

It is currently unknown how AOP is affected by alternations in body position. Therefore, the primary aim of the present study was to investigate the impact of body position on AOP by conducting a comparative investigation of lower body AOP determined by Doppler ultrasound in a supine vs. seated position using both a wide and a narrow cuff. Moreover, it is also unknown to which extent individual characteristics previously shown to predict AOP in a supine position are also predictive of AOP when seated. A secondary aim was therefore to investigate whether previously verified predictors of lower body AOP (i.e., thigh circumference and blood pressure) (Graham et al. 1993; Loenneke et al. 2015) can also serve as predictors of seated AOP. Collectively, these aims serve to increase the knowledge on how to improve standardization procedures on the degree of blood flow restriction in future studies on BFRE.

We hypothesized that AOP would increase when moving from a supine to seated position regardless of cuff width, with higher AOP when using a narrow cuff compared to a wide cuff. Furthermore, we hypothesized that thigh circumference and blood pressure can constitute predictors of AOP independently of body position.

## Methods

### Subjects

Initially, 24 healthy subjects (14 men and 10 women) were included in the study (see Table 1 for subject characteristics). All subjects were informed of the purpose and risks of the study and provided written informed consent in accordance with the standards of the Declaration of Helsinki. The study was conducted in accordance with ethical standards of the Ethical Committee for Region Midtjylland. Exclusion criteria were (1) cardiovascular or neural disease; (2) smoking; (3) pregnancy; (4) use of medication that could potentially influence the results.

### Study design

Prior to initiating the main study, we conducted a pilot study on seven subjects (not included in the main study) to determine the variation in AOP and allow sample size calculation. The pilot study showed that AOP was essentially identical (when reported as the nearest 5 mmHg) when replicate measurements were separated by 5 min on a given individual. Furthermore, the sample size was determined by completing a power analysis (power = 0.8, $\alpha = 0.05$) where correlation between supine and seated AOP was chosen as

**Table 1** Subject characteristics and arterial occlusion pressure

| Variable | Mean ± SD | 95% CI | Subject a (m) | Subject b (m) | Subject c (f) | Subject d (f) |
|---|---|---|---|---|---|---|
| Age (years) | 32 ± 13 | [26, 38] | 25 | 22 | 21 | 23 |
| Height (cm) | 178 ± 9 | [174, 182] | 184 | 182 | 166 | 172 |
| Weight (kg) | 73 ± 3 | [67, 79] | 90 | 98 | 73 | 79 |
| BMI (kg/m$^2$) | 23 ± 3 | [21.9, 24.4] | 26.5 | 29.7 | 26.3 | 26.7 |
| SBP (mmHg) | 116 ± 10 | [112, 121] | 139 | 134 | 134 | 117 |
| DBP (mmHg) | 71 ± 7 | [68, 75] | 70 | 80 | 77 | 83 |
| 33% circumference, *W* (cm) | 56.3 ± 3.7 | [54.6, 58.1] | 60.5 | 66.5 | 60.4 | 69.2 |
| 33% circumference, *N* (cm) | 56.1 ± 3.9 | [54.3, 57.0] | 60.5 | 63.5 | 57.4 | 69.2 |
| 33% skinfold (mm), *W* | 22.2 ± 11 | [17.1, 27.3] | 21.0 | 36.5 | 29.0 | 42.0 |
| 33% skinfold (mm), *N* | 22.4 ± 11 | [17.2, 27.6] | 23.2 | 36.5 | 25.1 | 47.8 |
| AOP supine, *W* (mmHg) | 148 ± 19 | [139, 157] | 170 | 170 | 150 | 225 |
| AOP seated, *W* (mmHg) | 177 ± 20** | [168, 186] | 215 | 230 | 175 | 260 |
| AOP supine, *N* (mmHg) | 344 ± 85$^{\#}$ | [304, 383] | 500 | – | – | – |
| AOP seated, *N* (mmHg) | 409 ± 101$^{*,\#}$ | [361, 456] | – | – | – | – |

Left side; mean values for subjects with an AOP < 600 mmHg ($n = 20$). Right side; individual values for subjects with an AOP > 600 mmHg. $^{*}p < 0.001$; $^{**}p < 0.0001$; significantly different from the supine position with the same cuff. $^{\#}p < 0.0001$; significantly different from the wide cuff in the same body position. No differences between thigh circumference or skinfold was observed between legs (wide leg vs. narrow leg)

*W* leg with wide cuff, *N* leg with narrow cuff, *m* male, *f* female, *BMI* body mass index, SBP systolic blood pressure, DBP diastolic blood pressure

the primary outcome. The analysis showed that a total of 20 subjects were required and we therefore included 24 subjects to account for potential dropouts and missing data.

The subjects were instructed to refrain from strenuous activity 24 h prior to visiting the laboratory and to allow a normal overnight sleep (i.e., 7–8 h of sleep). Moreover, they were instructed to refrain from consumption of caffeine 8 h and to refrain from dietary intake (except water) 2 h prior to visiting the laboratory. Upon arrival at the laboratory, the height, weight, thigh circumference and frontal thigh skinfold of each subject were measured, followed by 10 min of seated rest before measurements of the blood pressure. In randomized order, the subjects were then placed in the body position for the first determination of AOP (either supine or seated) and the wide cuff was positioned proximally on one leg and the narrow cuff on the contralateral leg (randomized for leg dominance). Following 5 min rest, AOP was first determined with either the wide or narrow cuff in randomized order after which an additional 5 min rest was given before determination of AOP with the other cuff on the contralateral leg. Finally, the body position was alternated for a second AOP determination and the subject rested for 5 min before determination of AOP with the wide and narrow cuff in randomized order separated by 5 min rest. AOP was determined twice for each leg (once for each body position) separated by a minimum of 5 min rest, which could theoretically influence the second AOP determination. However, this design was preferred over determination of AOP in different body positions on separate days, to eliminate influence of daily variation (Ingram et al. 2017). Furthermore, any influence of previous AOP determinations was most likely negated by the randomized order of tests used in the current study. Additionally, no order effects of AOP measurements were observed. All measurements (AOP, thigh circumference, skinfold, blood pressure) were performed by the same investigator with an intra-tester correlation of $R^2 \geq 0.98$.

## Thigh circumference and skinfold

The subjects were seated on the edge of a chair with hips and knees flexed to 90°. A mark was made on the frontal thigh distal to the inguinal crease at a distance corresponding to 33% of the distance from the inguinal crease to the top of patella, which was recorded using tape measure (approximate site of where cuffs would be applied). At the marked site, thigh circumference was first measured twice using tape measure. Subsequently, two measurements of thigh skinfold were conducted at the same site with a skinfold caliper (Harpenden Skinfold Caliper, Baty International, UK) in accordance with International Standards for Anthropometric Assessment (ISAK 2001) and the mean values were used for analysis.

## Blood pressure

Systolic blood pressure (SBP) and diastolic blood pressure (DBP) were measured using an automatic blood pressure cuff with digital display (Microlife Blood Pressure Monitor BP A100 Plus, Microlife, Widnau, Switzerland). The subjects had 10 min of seated rest before a minimum of two

measurements were taken for each subject. If SBP or DBP varied by more than 5 mmHg, measurements were repeated until values were within 5 mmHg of each other, whereupon the mean of the two closest values were averaged and used for analysis. Each measurement was separated by a minimum of 15 s rest.

### Determination of arterial occlusion pressure

The cuffs used for determining AOP were both obtained from the same manufacturer. The wide cuff was 13 cm wide with a 12-cm pneumatic bag inside (Hokanson, SC12; 13 × 85 cm, Bellevue, WA, USA) and the narrow cuff was 6 cm wide with a 5-cm pneumatic bag inside (Hokanson, SC5; 6 × 83 cm, Bellevue, WA, USA). Upon placing the subjects in the first body position for AOP determination, the wide and narrow cuffs were positioned proximally on each leg and the position was marked for accurate repositioning of the cuffs when the subject was moved to the second body position. Before inflation, cuffs were tightened to a snug fit and they were loosened immediately after determination of AOP to avoid any restriction of blood flow between AOP determinations. For determination of AOP, the pulse was initially detected using a hand-held Doppler probe (Dopplex D900, Huntleigh Healthcare Ltd., UK) placed on the posterior tibial artery. The cuff was then inflated to 70 mmHg using a digital tourniquet system with automatic regulation of pressure (Digital Tourniquet 9000, VBM, Medizintechnik, GmbH, Germany) and pressure was gradually increased in steps of 5 mmHg until auscultatory pulse was no longer present. The lowest cuff pressure at which the pulse was not present was considered the AOP. This procedure was repeated for both cuffs and body positions with 5 min rest assigned between AOP determinations. The tourniquet system used to inflate the cuffs had a built-in upper pressure limit of 600 mmHg. Consequently, if subjects still had detectable pulse when cuff pressure reached 600 mmHg, AOP was noted as "> 600 mmHg".

### Statistical analysis

AOP could not be determined in four of the subjects (two men and two women) with the narrow cuff (AOP > 600 mmHg). Therefore, to appropriately compare the influence of cuff width on AOP and allow for statistical analysis of paired data, datasets for these four subjects were omitted before analysis of the influence of body position and cuff width on AOP. However, for the secondary objective of the study to investigate if predictors of lower body AOP were similar in different body positions, data-points for these four subjects were used in regression analysis with regards to which variables may predict AOP with the wide cuff. This was done to strengthen our results to allow for better comparison to previous studies and because no direct comparison between wide and narrow cuffs were intended.

Data were checked for normality of distribution by Shapiro–Wilk test and QQ plots. Differences in anthropometric measures between legs (wide cuff leg vs. narrow cuff leg) were assessed by paired *t* tests. Paired *t* tests were employed to determine if AOP increased when moving from a supine to seated position and if AOP was higher with the narrow cuff compared to the wide cuff. Effect sizes for the magnitude of differences in AOP between body positions and cuff widths were determined using Cohen's *d* and are reported as means and 95% CI. To determine if seated AOP would depend on supine AOP and quantify the relationship, linear regression analysis was used. A multiple linear regression analysis including variables; thigh circumference, skinfold, DBP and SBP, was performed to assess which variables that constitute predictors AOP and whether these are independent of body position. Following this, to generate prediction equations for AOP, regression analysis was repeated with only those variables found to significantly predict AOP. Multicollinearity between variables was defined as a VIF ≥ 10 and/or Pearson correlations ≥ 0.85.

Alpha level was set to $p \leq 0.05$ and data are presented as mean ± SD. Statistical analysis was made in Stata 12.0 (Statacorp, College Station, TX, USA) and graphical presentations were made in GraphPad version 7.0 (GraphPad Software, La Jolla, CA, USA).

## Results

Subject characteristics are shown on the left side of Table 1, with characteristics for subjects exhibiting AOP < 600 mmHg with the narrow cuff shown on the right side of Table 1. No differences between legs were observed in anthropometric measures.

### Effect of body position and cuff width on arterial occlusion pressure

AOP increased from the supine to seated position with both cuff sizes and AOP was higher with the narrow cuff compared to the wide cuff in both body positions. In accordance, in the supine position, AOP reached 148 ± 19 and 348 ± 85 mmHg while in the seated position, AOP increased to 177 ± 20 and 409 ± 101 mmHg, with the wide and narrow cuff, respectively. This corresponded to differences of 29 ± 9 and 65 ± 73 mmHg between the wide and narrow cuff (Table 1; Fig. 1). Effect sizes for the magnitude of changes in AOP when altering the body position were 1.5 [0.8, 2.2] and 0.7 [0.05, 1.3] for the wide and narrow cuff, respectively, and effect sizes for the difference in AOP between the wide and narrow cuff were 3.2 [2.2, 4.1] in both body positions.

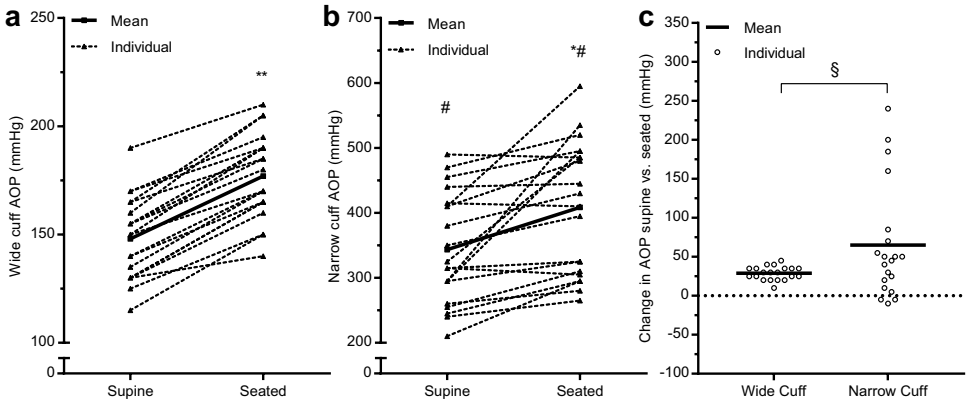


**Fig. 1** Effect of body position on AOP with a **a** wide and **b** narrow cuff ($n=20$). Dashed and solid lines represent individual and mean values, respectively. **c** Individual and mean differences between supine and seated AOP. Circles indicate individual changes and solid lines indicate mean changes. $^{*}p<0.01$; $^{**}p<0.0001$; significantly different from the supine position with the same cuff. $^{\#}p<0.0001$; significantly different from the wide cuff in the same body position. $^{\S}p<0.05$; difference between cuffs

A positive correlation between supine and seated AOP was observed with both the wide ($R^2=0.81$, $p<0.0001$) and narrow cuff ($R^2=0.50$, $p<0.001$) (Fig. 2).

No differences between genders were observed when the analysis of the influence of body position on AOP was performed separately in men and women (supplemental material, Table 1).

## Determinants of lower body arterial occlusion pressure

Results of multiple regression analysis performed on 24 and 20 subjects with the wide and narrow cuffs, respectively, are shown in Table 2. None of the variables showed multicollinearity.

For the wide cuff, the standardized coefficients indicated that thigh circumference explained most of the variance in AOP with DBP explaining additional variance in both body positions. For the narrow cuff, only thigh circumference served as predictor of AOP. Skinfold and SBP did not predict of AOP in any condition.

Repeating the regression analysis for the wide cuff after omitting the four subjects who exhibited AOPs > 600 mmHg, reduced the overall adj. $R^2$ (supine 0.381, seated 0.461). Only thigh circumference was observed to constitute a predictor of AOP (data not shown), thereby resembling the results from regression analysis with the narrow cuff. Results from regression analysis and prediction equations including only variables with significant impact on AOP (wide cuff; thigh circumference and DBP, narrow cuff; thigh circumference) are shown in Table 3.

## Discussion

The main findings of the study were (1) that AOP is influenced by body position; (2) that a much higher pressure is necessary to reach AOP when using a narrow cuff compared

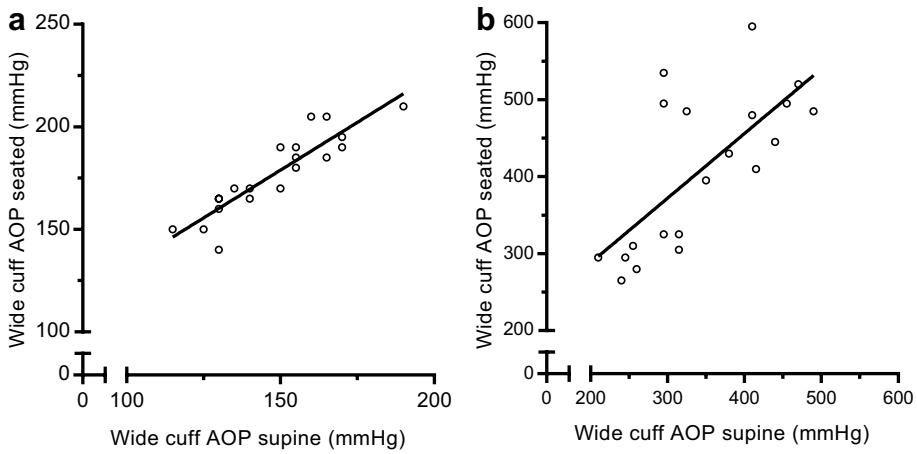


**Fig. 2** Linear regression of supine and seated AOP with a **a** wide ($R^2=0.81$, $p<0.0001$) and **b** narrow cuff ($R^2=0.50$, $p<0.001$) ($n=20$). NB; two subjects had identical AOPs with the wide cuff and thus only 19 individual values are visible in **a**

**Table 2** Multiple regression analysis of variables influencing arterial occlusions pressure (wide cuff $n = 24$; narrow cuff $n = 20$)

| Variable | Stand. coef. | $p$ (variable) | $R^2$ | Adj. $R^2$ | $p$ (model) | Root MSE |
|---|---|---|---|---|---|---|
| Wide supine | | | | | | |
| 33% circumference | 0.525 | < 0.01 | 0.631 | 0.553 | < 0.001 | 15.935 |
| 33% skinfold | 0.204 | 0.230 | | | | |
| DBP | 0.379 | < 0.05 | | | | |
| SBP | − 0.002 | 0.989 | | | | |
| Wide seated | | | | | | |
| 33% circumference | 0.572 | < 0.001 | 0.723 | 0.655 | < 0.0001 | 15.844 |
| 33% skinfold | 0.217 | 0.143 | | | | |
| DBP | 0.318 | < 0.05 | | | | |
| SBP | 0.097 | 0.547 | | | | |
| Narrow supine | | | | | | |
| 33% circumference | 0.604 | < 0.01 | 0.438 | 0.288 | 0.057 | 71.529 |
| 33% skinfold | 0.199 | 0.388 | | | | |
| DBP | 0.351 | 0.225 | | | | |
| SBP | − 0.183 | 0.535 | | | | |
| Narrow seated | | | | | | |
| 33% circumference | 0.629 | < 0.01 | 0.420 | 0.265 | 0.070 | 86.652 |
| 33% skinfold | − 0.136 | 0.561 | | | | |
| DBP | − 0.130 | 0.652 | | | | |
| SBP | − 0.132 | 0.659 | | | | |

$p$ (model) depicts $p$ value for the overall regression model; $p$ (variable) depicts $p$ value for independent variable coefficients

*SBP* systolic blood pressure, *DBP* diastolic blood pressure, *MSE* mean square error

**Table 3** Regression analysis and prediction equations including only variables with significant influence on AOP

| | $R^2$ | Adj. $R^2$ | $p$ | Root MSE | Prediction equation |
|---|---|---|---|---|---|
| Wide supine | 0.600 | 0.561 | < 0.0001 | 15.800 | AOP (mmHg) = 3.105 (thigh circumference) + 1.221 (DBP) − 114.015 |
| Wide seated | 0.688 | 0.659 | < 0.0001 | 14.992 | AOP (mmHg) = 3.925 (thigh circumference) + 1.386 (DBP) − 142.166 |
| Narrow supine | 0.376 | 0.302 | < 0.05 | 70.818 | AOP (mmHg) = 13.063 (thigh circumference) − 389.920 |
| Narrow seated | 0.375 | 0.340 | < 0.01 | 82.139 | AOP (mmHg) = 15.976 (thigh circumference) − 488.488 |

Thigh circumference (cm) and DBP (mmHg) were included in regression analysis of AOP with wide cuff and thigh circumference were included with the narrow cuff

*AOP* arterial occlusion pressure, *DBP* diastolic blood pressure, *MSE* mean square error

to a wide cuff; and (3) that a strong relationship exist between supine and seated AOP. Also noteworthy, very high levels (> 500–600 mmHg) may be required to achieve AOP with a narrow cuff in certain individuals. This information is highly valuable for standardization of the degree of blood flow restriction levels during BFRE.

## Influence of body position using different cuff widths on arterial occlusion pressure

The use of individually set relative cuff pressures (% of AOP) has been proposed as a means to ensure a uniform degree of blood flow restriction between individuals engaging in BFRE studies (Loenneke et al. 2014; Mouser et al. 2017). Yet, determination of the relative cuff pressure used during exercise is often based on measurement of AOP in the supine position (Laurentino et al. 2012; Fahs et al. 2015; Lixandrão et al. 2015). We considered that this does not take the effect of hydrostatic pressure on AOP during seated BFRE into account. In accordance, in the current study, we demonstrate that lower limb AOP increases when moving from a supine to seated position, in a manner resembling the expected added hydrostatic pressure (Folkow et al. 1971; Eiken 1988). In absolute terms, AOP was observed to increase more with the narrow cuff than the wide cuff (65 vs. 29 mmHg), while the mean relative changes were almost identical (20.0 ± 6.9 and 20.8 ± 23.7% for the wide and narrow cuff, respectively). The greater absolute pressure

increase with the narrow cuff is assumed to emerge due to the large absolute difference in supine AOP between the wide and narrow cuff (148 vs. 344 mmHg, respectively). Furthermore, we observed a strong relationship between supine and seated AOP with the wide cuff, whereas with the narrow cuff, greater variation between individuals reduced the strength of the relationship.

Collectively, our results demonstrate that prescribing relative cuff pressures based on AOP measured in a supine position before performing lower body BFR exercise in a seated position may result in overestimation of the relative blood flow restriction. This is important to take into account in standardization procedures in future studies on the effects of BFRE. Whether the overestimation of blood flow restriction will have great impact on, e.g., muscle adaptations remains speculative, as acute (Loenneke et al. 2016) as well as prolonged training studies (Lixandrão et al. 2015) using lower body BFRE, have shown that pressures as low as 40% of supine AOP can impose a sufficient stimulus for muscle growth, and seated exercise at 50% of supine AOP would still correspond to exercise at ~40% of seated AOP.

## Similarities and differences in results compared to previous studies

The higher pressure required to occlude arterial blood flow when using the narrow cuff is in agreement with previous studies comparing AOP with wide vs. narrow cuffs (Crenshaw et al. 1988; Loenneke et al. 2012). Accordingly, Loenneke et al. (2012) observed a mean supine AOP with a wide cuff (identical brand and model as the wide cuff used in our study) of 144 mmHg which is comparable to the 148 mmHg found in the current study. On the other hand, when using a narrow cuff, Loenneke et al. (2012) found a mean supine AOP of 235 mmHg while 344 mmHg was observed in the current study. The higher AOP observed by us is likely explained by the fact that the Hokanson E20 rapid cuff inflator used by Loenneke et al. (2012) can only increase pressures up to 300 mmHg, while the digital tourniquet employed by us, allowed increases in pressure up to 600 mmHg. Similarly, Crenshaw et al. (1988) reported an upper limit of 360 mmHg in their measurement system, and as several of their subjects in that study exhibited AOP levels at or above this limit with a narrow 4.5 cm cuff, it is possible that the AOP may have been underestimated in some of their subjects. Consequently, our results therefore confirm the results from these previous studies. Furthermore, we considerably extend the range of pressures required to achieve AOP with narrow pneumatic cuffs.

Another difference between our study and Loenneke et al. (2012) is that they used a 5-cm wide Kaatsu Master cuff as the narrow cuff, while we used a Hokanson SC5 cuff. However, in a follow-up study by Loenneke et al. (2013), it was demonstrated that the Kaatsu Master and the Hokanson SC5 cuff yielded similar AOPs (~240 mmHg) in subjects for whom it was possible to determine AOP in the proximal thigh (Loenneke et al. 2013). However, for about half of the subjects in that study, it was not possible to determine AOP with either cuff (six subjects had > 300 mmHg with the Kaatsu Master and eight subjects had > 300 mmHg with the SC5). Yet, the data indicated that the 5-cm Kaatsu Master cuff and the Hokanson SC5 cuff yielded similar AOPs. In other words, the models of narrow cuffs are likely not the reason for the different AOPs observed.

The high overall mean values of AOP with the narrow cuff, and in particular the extreme AOPs (> 500–600 mmHg) that we observed in some subjects, may be surprising at a first glance. However, similarly high levels of AOP have previously been reported in studies using non-pneumatic 3.8 cm wide cuffs placed at the mid-thigh level (Wall et al. 2013, 2015). Moreover, a study by Graham et al. (1993) reported AOPs at or above 400 mmHg when the ratio between cuff width and limb circumference was less than 0.1:1. In our study, the four subjects which exhibited AOPs over 600 mmHg had thigh circumferences between 57.4 and 69.2 cm (Table 1), resulting in cuff width to limb circumference ratios of between 0.07:1 and 0.09:1, considering that the bladder width (the effective part of the cuff) was 5 cm for the SC5 cuff. Thus, our results are in good agreement with those of Graham et al. (1993) and Wall et al. (2013, 2015).

## Implications for blood flow restricted exercise

The substantial inter-individual variation in AOP observed for the narrow cuff is noteworthy (i.e., AOP exceeded 300 mmHg in 12 and 16 subjects out of 20 subjects in the supine and seated position, respectively). Consequently, previous BFRE studies that have employed narrow cuffs combined with fixed absolute cuff pressures (typically in the range of 160–240 mmHg) (Karabulut et al. 2011) or pressures based on systolic blood pressure (typically 130% above SBP) (Cook et al. 2007; Clark et al. 2011) have entailed lack of control of exact degree of relative blood flow restriction between individuals. Moreover, our results strongly suggest that even when the BFR pressure is set as a percentage of AOP, certain variability in the degree of occlusion is introduced whenever the AOP is determined in a posture different from that employed during BFR exercise, especially if narrow cuffs are used.

In addition, it has been hypothesized that a too high relative cuff pressure may increase the health risks (Loenneke et al. 2014). In the current study, four subjects exhibited AOPs > 600 mmHg. These specific individuals were also characterized by exhibiting some of the highest values in BMI, skinfold, thigh circumference and/or blood pressure (Table 1). For BFRE to be eligible in the clinical settings

(Hughes et al. 2017), the use of wider cuffs may therefore be preferred. Moreover, under such circumstances it may not be possible to use relative cuff pressures based on AOP with narrow cuffs, as many tourniquet systems are inherent of upper limits of 300–400 mmHg, thus making determination of true AOP impossible in many subjects.

Whether BFRE is to be employed on healthy or patient populations, it is also relevant to consider the potential driving effects of factors related to fatigue vs. other potential drivers on muscle accretion. In accordance, in a previous study, we provided evidence that low-load BFRE and low-load free-flow resistance exercise, were equally capable of producing muscle hypertrophy, provided that both regimes were conducted to volitional failure (Farup et al. 2015). However, other studies employ standardized repetition schemes (typically 30–15–15–15). The extent to which fixed repetition schemes are fatiguing is obviously dependent on the degree blood flow restriction. Under such conditions, lack of control of AOP entail the risk that some individuals will experience exhaustion (perhaps even precluding them from completing all repetitions), whereas other individuals would need to perform additional work to reach exhaustion, thus probably resulting in marked differences in the degree of muscle fiber recruitment between individuals. This approach therefore entails the potential risk that the effect on muscle adaptations of a BFRE training regime cannot be precisely judged. Consequently, we argue that especially fixed repetition BFRE protocols combined with use of narrow cuffs and alternations in body position would benefit from taking the information gathered in the current study into account.

### Prediction of lower body arterial occlusion pressure

Multiple linear regression was used to determine if easily measured variables of influence on AOP, such as thigh circumference, skinfold, DBP and SBP could serve as predictors of AOP. This was evaluated on data gathered from both a supine and seated body position with both wide and narrow cuffs. Prediction equations including variables with impact on AOP are presented in Table 3. In the previous study by Loenneke et al. (2012), using an identical wide cuff, it was found that thigh circumference, ankle brachial index (ABI) and diastolic blood pressure (DBP) could serve as predictors of supine AOP (Loenneke et al. 2012). Although we did not determine ABI, our results are in agreement that AOP can be predicted by thigh circumference and DBP, whereas systolic blood pressure (SBP) and thigh composition (skinfold) does not seem to constitute strong predictors (Crenshaw et al. 1988; Loenneke et al. 2012). In addition, the strength of the predictors of AOP did not change when AOP was determined in the seated position (Table 2). Correspondingly, when comparing standardized coefficients from the supine and seated position in our study to the supine position in Loenneke et al. (2012), they appear quite similar. This indicates that, despite an increased absolute AOP in the seated position, the impact of each variable influencing AOP is held constant. Further support of this notion emerges from the observation of a strong correlation between supine and seated AOP.

However, when using a narrow cuff, we only found thigh circumference to serve as a predictor of both supine and seated AOP. This is in contrast to previous studies that have found DBP (Loenneke et al. 2012) and SBP (Loenneke et al. 2015) to constitute predictors of AOP in a supine position (although with lesser predictive strength than thigh circumference). We acknowledge that, compared to the aforementioned studies, we were likely underpowered to strongly conclude on the impact of DBP or SBP with the narrow cuff. Alternatively, since an upper detection limit of 600 mmHg was used in our study as opposed to only 300 mmHg in the studies by Loenneke et al. (2012, 2015), another explanation may be that the impact of DBP or SBP decreases with pressures > 300 mmHg. Regardless, previous findings combined with our current results indicate that < 50% of the variance in AOP is explained by individual characteristics with narrow cuffs compared to 55–66% when using wide cuffs (Loenneke et al. 2012, 2015).

## Limitations

We acknowledge that our study inherently has some limitations. First, it is important to emphasize that the results are only telling of the influence of factors on AOP measurements for lower body BFRE, whereas they cannot readily be translated to upper body BFRE. The influence of body position on AOP for upper body exercise is likely considerably smaller as the near proximity of the upper extremities to the heart confers less influence of hydrostatic pressure. Furthermore, we did not investigate the effect of a standing position. However, with regards to BFRE conducted as whole-body exercise (e.g., BFR squat exercise), taking hydrostatic pressure into account to determine AOP, is likely even more necessary, as the distance to the heart increases even further than with seated BFRE. Second, AOP was determined in a resting state for standardization purposes and performing exercise will in itself likely increase the pressure required to obtain AOP. In principle, it is therefore also necessary to account for exercise-induced changes in blood pressure to determine AOP with the utmost accuracy, but this may prove very difficult. Third, our prediction equations provided in Table 3 are only valid for the specific 12- and 5-cm cuffs employed, respectively, and may therefore not provide an accurate estimate of AOP with other cuff widths. Finally, it

is currently unknown whether the results translate directly to clinical populations.

## Conclusion

In conclusion, the current study supports our hypothesis that body position exerts substantial influence on absolute cuff pressures required to obtain arterial occlusion. Furthermore, the differences between narrow and wide cuffs in arterial occlusion pressures appear to be substantial, and greater than what has been reported previously. Importantly, very high levels of pressure may be necessary to reach arterial occlusion with narrow cuffs in some individuals, particularly those with large limb circumferences. These findings should be considered to be implemented in standardization procedures for controlling the relative degree of blood flow restriction during BFRE, especially if performed with a fixed repetition scheme. Finally, our study confirms the ability of thigh circumference and DBP to serve as predictors of AOP irrespective of body position.

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### Compliance with ethical standards

**Conflict of interest** The authors declare that they have no conflict of interest.

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